

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Group No: 12

Team Members:

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1. Problem Title:

Contact Management System

STRUCTURAL DESIGN

2. Class diagram

(Draw class diagram that shows different classes and their relationships. Use UML notations)

Classes:

- **User** – userID, username, password
- **Contact** – contactID, name, email, address
- **PhoneNumber** – numberID, phoneNumber
- **Group** – groupID, groupName

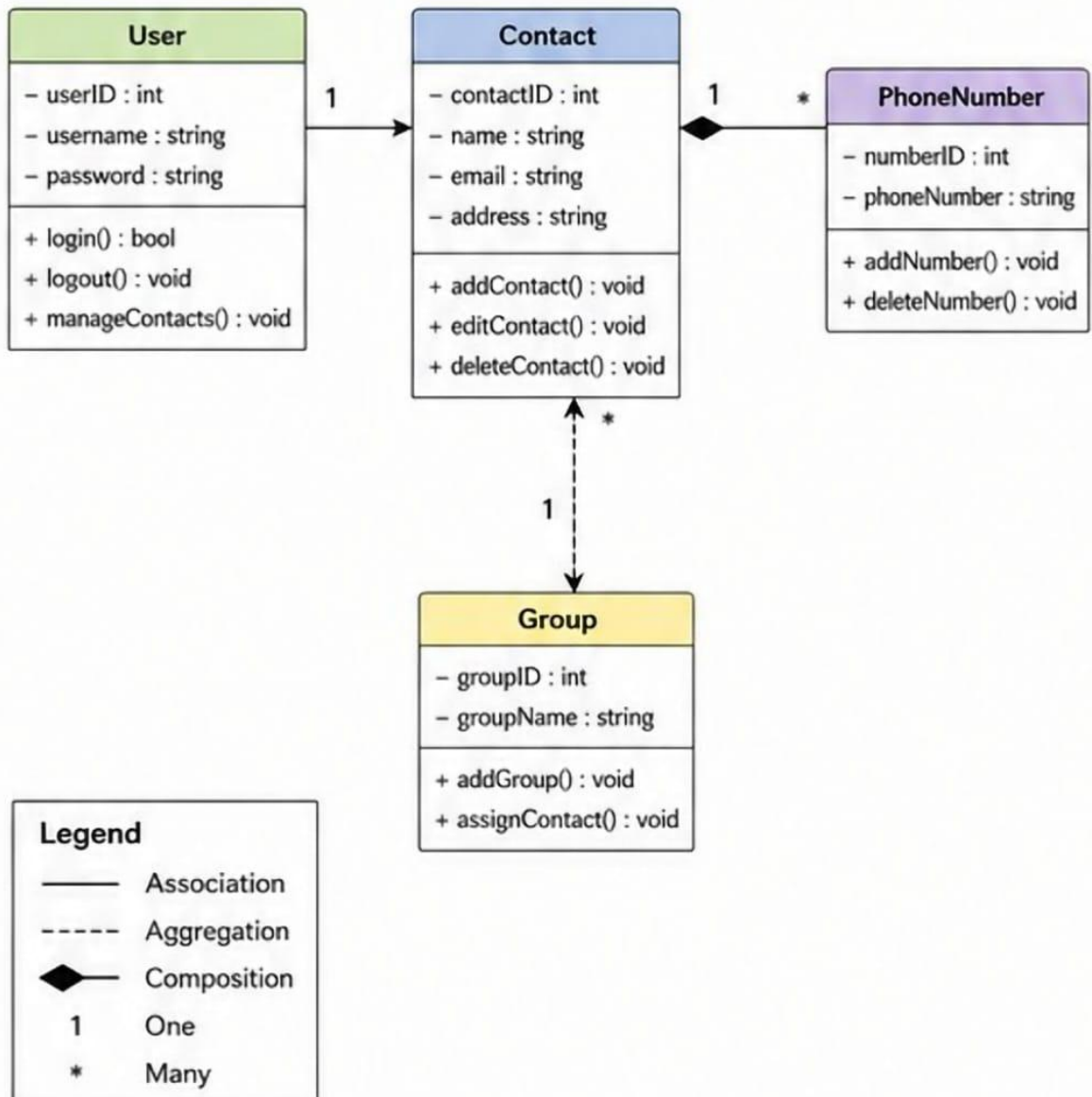
Relationships:

- User → Contact (1..*)

- Contact → PhoneNumber (1..*)
- Contact → Group (many-to-many)

STRUCTURAL DESIGN

Class diagram (draw class diagram that shows different classes and their relationships. Use UML notations)



BEHAVIOURAL DESIGN

3. Use-case diagram

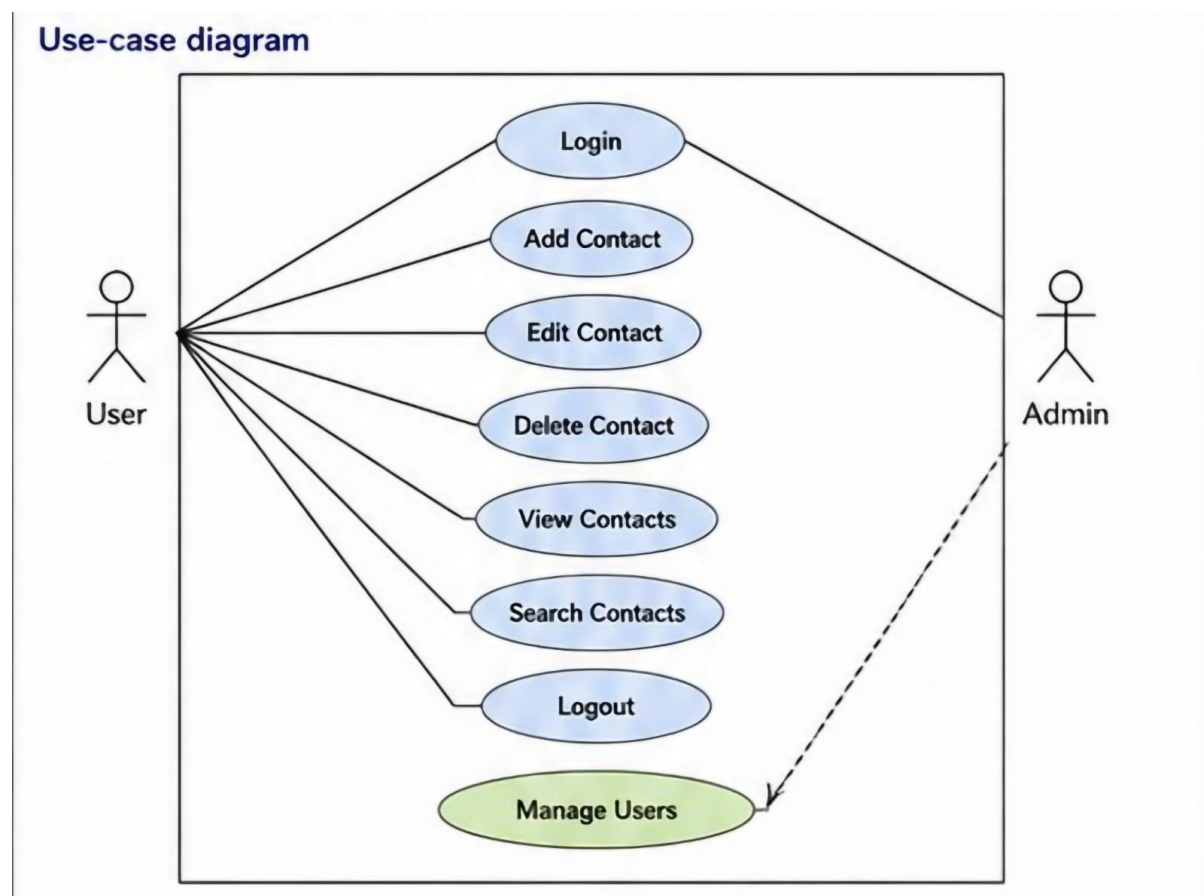
a. List of actors

1. User – Manages contacts
2. Admin – Manages users and system

b. List of use-cases

1. **Login** – User enters credentials and accesses the system.
2. **Logout** – User exits the system securely.
3. **Add Contact** – User adds new contact details.
4. **Edit Contact** – User updates existing contact details.
5. **Delete Contact** – User removes a contact.
6. **View Contacts** – User views all contacts.
7. **Search Contacts** – User searches contacts using keywords.
8. **Manage Users** – Admin manages system users.

c. Use case diagram

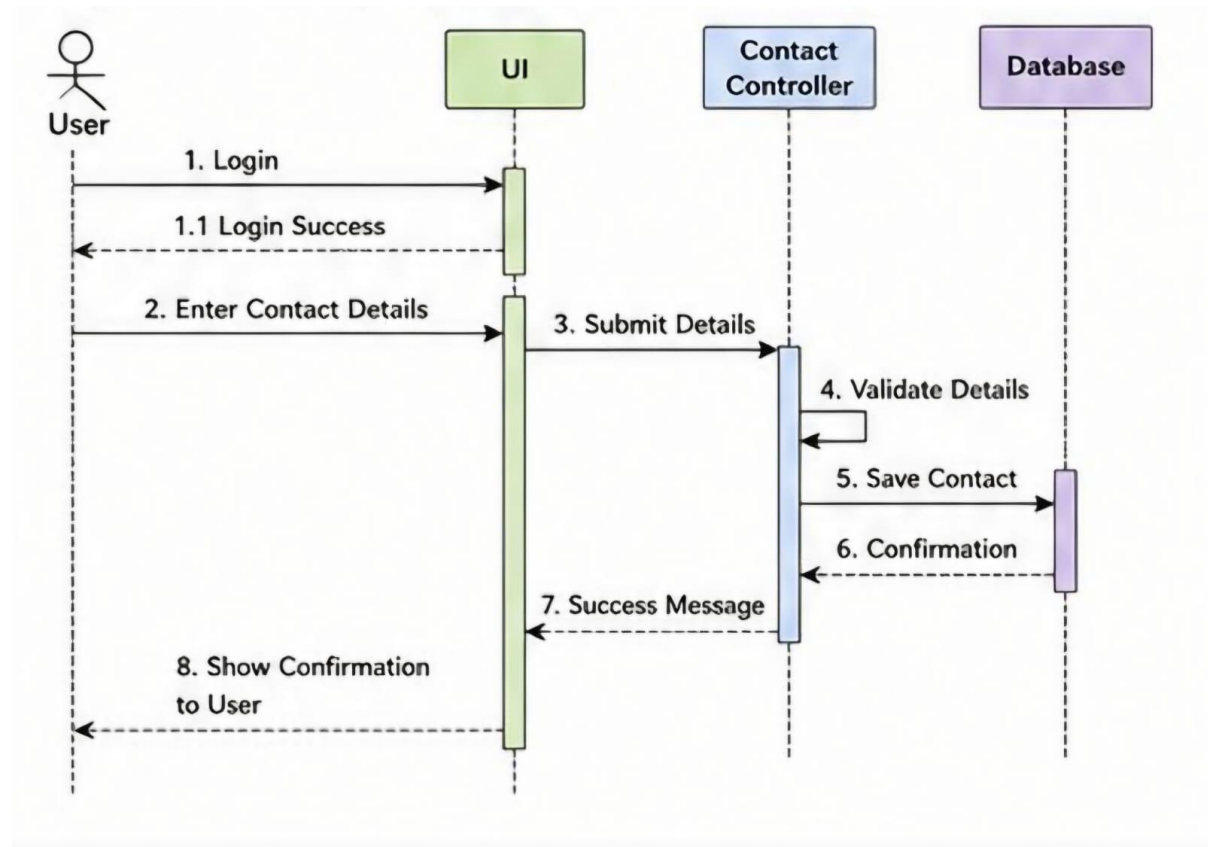


4. Sequence diagram (use UML notations)

a. Use Case: Add Contact

Scenario:

The user logs into the system and enters contact details. The system validates the input, stores it in the database, and displays a confirmation message.



b. Use Case: Search Contact

Scenario:

The user enters a keyword to search for a contact. The system processes the request, retrieves matching contacts from the database, and displays the results.

